

PRACTICE TEST 2

1. What does a systems programmer do? Write a short paragraph.

2. Read the following email and answer the questions below.

Dear Ms Jones,

You asked in the meeting whether we should purchase Mac® computers or PCs that run Microsoft® Windows® or Linux®. I feel that we would be better off investing in Mac® computers.

We are currently using PCs and have had many issues with them. I have to upgrade the Windows® operating system with patches, which is time-consuming. While PCs are highly software compatible, they are also more vulnerable to attack, so we need expensive security software.

One advantage to PCs is the vast availability of hardware choices. It might be easier to customise the computers to meet individual employee needs.

But Macs® still meet our needs better than PCs. The Mac OS X® needs little maintenance. Since our company develops architectural designs, we need computers with good graphics. Apple® has outstanding prepress and colour matching capability, as well as great processing speed.

One main drawback of replacing our current PCs with Macs® is that of training. Currently, all of our employees use the Windows® operating system. If we purchase Apple® computers, we would need to teach our employees how to use OS X®.

I hope this helps guide your decision.

Yours sincerely,

Simon Jacobs
IT support

a. What does the man recommend and why?

b. Use the email and write a new one about operating systems. Include your recommendation.

3. Write a descriptive text based on the information below.

	Linux
type	Unix-based
computer	wide variety
features	variety of distribution kits available
graphics engine	XFree86
User interface type	command line, GUI
User interface	KDE, Gnome
Source code availability	freely available

4. Read the text and answer the questions below.

LINUX



Linux has its roots in a student project. In 1992, an undergraduate called Linus Torvalds was studying computer science in Helsinki, Finland. Like most computer science courses, a big component of it was taught on (and about) Unix. Unix was the wonder operating system of the 1970s and 1980s: both a textbook example of the principles of operating system design, and sufficiently robust to be the standard OS in engineering and scientific computing. But Unix was a commercial product (licensed by AT&T to a number of resellers), and cost more than a student could pay.

Annoyed by the shortcomings of Minix (a compact Unix clone written as a teaching aid by Professor Andy Tannenbaum) Linus set out to write his own 'kernel' – the core of an operating system that handles memory allocation, talks to hardware devices, and makes sure everything keeps running. He used the GNU programming tools developed by Richard Stallman's Free Software Foundation, an organisation of volunteers dedicated to fulfilling Stallman's ideal of making good software that anyone could use without paying. When he'd written a basic kernel, he released the source code to the Linux kernel on the Internet.

Source code is important. It's the original from which compiled programs are generated. If you don't have the source code to a program, you can't modify it to fix bugs or add new features. Most software companies won't sell you their source code, or will only do so for an eye-watering price, because they believe that if they

make it available it will destroy their revenue stream.

What happened next was astounding, from the conventional, commercial software industry point of view – and utterly predictable to anyone who knew about the Free Software Foundation. Programmers (mostly academics and students) began using Linux. They found that it didn't do things they wanted it to do – so they fixed it. And where they improved it, they sent the improvements to Linus, who rolled them into the kernel. And Linux began to grow.

There's a term for this model of software development; it's called Open Source (see www.opensource.org/ for more information). Anyone can have the source code – it's free (in the sense of free speech, not free beer). Anyone can contribute to it. If you use it heavily you may want to extend or develop or fix bugs in it – and it is so easy to give your fixes back to the community that most people do so.

An operating system kernel on its own isn't a lot of use; but Linux was purposefully designed as a near-clone of Unix, and there is a lot of software out there that is free and was designed to compile on Linux. By about 1992, the first 'distributions' appeared.

A distribution is the Linux-user term for a complete operating system kit, complete with the utilities and applications you need to make it do useful things – command interpreters, programming tools, text editors, typesetting tools, and graphical user interfaces based on the X windowing system. X is a standard in academic and scientific computing, but not hitherto common on PCs; it's a complex distributed windowing system on which people implement graphical interfaces like KDE and Gnome.

As more and more people got to know about Linux, some of them began to port the Linux kernel to run on non-standard computers. Because it's free, Linux is now the most widely-ported operating system there is.

- a. What did Linus Torvalds use to write the Linux kernel?
- b. What type of utilities and applications are provided in a Linux distribution? (
- c. What is X?
- d. What graphical user interfaces are mentioned in the text?

5. Mark the following statements as True or False.

- a. Linux was created in the 1980s. _____
- b. Minix was created by a university student. _____
- c. Linux is based on Unix _____
- d. Minix is based on Unix. _____
- e. Linux runs on more types of computer than any other operating system. _____

6. GUI. Match the interface elements with the following definitions.

- a) window _____ a menu that the user 'pulls down' from a name in the menu bar at the top of the screen by selecting the name with the mouse
- b) dock _____ graphic images (or intuitive symbols) used to represent an object or task
- c) tool bar buttons _____ containers for documents and applications
- d) folders _____ found at the top of a window, they take you to the Home folder and others.
- e) icons _____ set of icons at the bottom of the screen that give you instant access to the things you use most
- f) pull-down menu _____ an arrow, controlled by the mouse, that allows you to move around the screen
- g) the pointer _____ a viewing area less than or equal to the screen size. By using different windows you can work on several documents or applications simultaneously

7. Correct the following statements

- a) LANs link computers and other devices that are placed far apart.
- b) The word protocol refers to the shape of the network.

- c) The Internet is an example of a LAN.
- d) A group of stars connected to a central bus is called a ring.

8. Fill in the gaps with words from the box.

digital certificate malware virus spyware firewall antivirus

Malicious software, _____, can be avoided by following some basic rules:

Internet users who like cybershopping should get a _____, an electronic identity card.

To prevent hackers from breaking into your internal network and obtaining your data, install a _____.

It will protect you from _____.

If you have been hit by a _____, don't panic. Download a clean-up utility and always remember to use an _____.

9. Read the text below and complete it with the correct form of the words in brackets.

A Brief History of the iOS App Store and the Rise of the Mobile App

One of the biggest things to come out of the smartphone revolution is the rise of the mobile app. Before Apple, Google and other companies set to create their own on-device app stores _____ (use) were left to live a life of boring apps that needed _____ (install) via a memory stick or, in the case of some smartphone operating systems, the downloading of _____ (execute) files that needed to be installed _____ (manual) after going through a selection of _____ (secure) issues.

Apple, along with Research in Motion, began to change all that with a little help from Google. Nokia also got in on the act, before Microsoft finally began to get things right. Apps, as we have all learned over the last few years, sell smartphones. Now, they also sell tablets.

Back in 2007 most people didn't own smartphones. They were thought of as being expensive and limited to doing what other phones did – make calls and let people send SMS.

However, after _____ (release) the iPod touch, Apple and its then-CEO, Steve Jobs, started showing interest in web apps, believing they were the future. A few years later they launched Apple's App Store.

There are now one million apps available across a variety of app stores from all kinds of _____ (vend).

According to recent studies, _____ (develop) seem to understand that the key to getting people to use their apps is to make them _____ (afford), or even free. Besides _____

(advertise), a free app can become _____ (profit) if enough people download it. In-app

_____ (purchase) can sometimes earn just as much, if not more, than charging a flat rate for an app.

Apps are the next _____ (logic) step for a worldwide web that is becoming less of a home for text and more of a place where people go to interact, watch, and listen to media.

10. Read the text and put the steps in the correct order to make a flowchart showing how this type of VoIP phone system works.

How VoIP phone systems work

VoIP (Voice over Internet Protocol) phone systems work by sending data via the Internet in tiny packets. This is called packet switching. It works like this:

- 1 Your voice signal, which is analogue, is converted into digital data. If you have a standard phone, you need an extra piece of hardware to do this called an ATA (Analogue telephone adaptor). If you have an IP phone, it produces a digital signal so you don't need an adaptor. You can also use your PC and a microphone as a telephone.
- 2 The sending computer uses software to compress the digital data, much like MP3 files.
- 3 The data is divided into packets, each one 30 milliseconds long.
- 4 The packets are sent to a router which decides the best path through the Internet for each packet. They will travel by many different paths. They will arrive at different times and some may even be lost.
- 5 The receiving computer uses special software to store the packets and put them in the right order. Because the packets are so small, you won't hear the difference if some are lost.
- 6 The data is converted back to voice and played through your standard phone, IP phone, or PC headphones.

If you have a wireless VoIP handset, you can make and receive calls anywhere near a wireless hotspot. Some mobile phones are dual-mode. You can use a mobile phone network or wireless VoIP, depending where you are.

- a) The packets are sent to a router.
- b) The digital data is compressed by the sending computer.
- c) The data is converted back to voice.
- d) The receiving computer puts the packets back together again.
- e) The voice signal is converted to digital.
- f) The digital data is divided into very small packets.
- g) The router sends each packet through the Internet by the best available path.

11. Rewrite the following sentences using the present passive.

- a. First, the technician turns on the computer and accesses the start-up program.
- b. He runs the scan disk utility software and checks the files.
- c. The user inserts the password.
- d. They send the files to central office.

12. Complete the CV with these headings.

- | | |
|---|--|
| a. Education and training | f. Position held |
| b. Main activities and responsibilities | g. Subjects/ Occupational skills covered |
| c. Organisational details | h. Technical skills |
| d. Personal information | i. Title of qualifications awarded |
| e. Personal skills | j. Work experience |



Europass Curriculum Vitae

(1) _____

First name(s) / Surname(s)

Sukvinder Dhal

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Telephone(s)

+ (44) 1982 12459

E-mail

skvdhl8618@coolmail.com

(2) _____

Dates

2011–present

(3) _____

IT Support Technician

(4) _____

Provide support to customers of my company's networking products, both wireless and wired. Diagnose problems and suggest solutions.

Name and address of employer

Agoda Digital Ltd, Garside Industrial Estate, Sunderland, UK

(5) _____

Dates

2009–2011

(6) _____

BTEC National Diploma for IT Practitioners (Systems Support)

(7) _____

IT technical support, customer service, networking (wireless and wired), hardware installation, Conglefield College, Cheshire, UK

Name and type
of organisation providing
education and training

Personal skills and competences

Mother tongue(s)

Hindi

(8) _____

I am a good team player at work. I work well with people from many different backgrounds. People find me friendly and easy to work with and I understand other people easily. I enjoy helping customers.

(9) _____

I often prepare my team's schedules. When problems happen, I often make useful suggestions; I have good problem-solving skills. I am very organised; this helps me to work efficiently. I enjoy taking responsibility.

(10) _____

Setting up both wired and wireless networks, including Wi-Fi. Diagnosing problems in networks. Server administration of both Linux and Windows operating systems.